On the Identifiability of Latent Romantic Interest: A Sequential Hypothesis Testing and Bayesian Filtering Approach to Determining Whether She Likes You

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Abstract

We study the problem of inferring a latent binary parameter $\theta \in \{0, 1\}$, the indicator that she likes you, from a sequence of noisy behavioral observations taking values in a finite *signal alphabet*. We first establish that θ is identifiable if and only if the two conditional signal laws are distinct, and prove a non-identifiability theorem for the case in which she is *just being nice*. Under identifiability, the posterior log-odds evolve as a random walk whose drift is a Kullback–Leibler divergence, and we derive the optimal stopping rule via Wald's Sequential Probability Ratio Test, obtaining that approximately 11 interactions are required in expectation if she does and 16 if she does not, at error rates of 5%. We then show that an observer who overweights favorable signals by a factor exceeding an explicit critical value $\kappa^* \approx 2.28$ converges almost surely to the belief that she likes you, *regardless of the truth*. For time-varying interest modeled as a latent Gaussian AR(1) process, the steady-state Kalman filter error is bounded away from zero by the positive root of a discrete algebraic Riccati equation, so certainty is never attained. Finally, a value-of-information argument identifies the optimal experiment: asking her.

1 Preliminaries and the Signal Space

Let $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space, where $\mathscr{F}_t$ is the σ-algebra generated by all interactions observed up to time t (texts, replies, eye contact in the Reg, whether she laughed at the joke that was not funny).

Definition 1.1 (Latent Interest)**.** The *latent interest* is an $\mathscr{F}$-measurable random variable $\theta : \Omega \to \{0, 1\}$, where $\{\theta = 1\}$ is the event that she likes you. Its law is the *prior* $\pi_0 = \mathbb{P}(\theta = 1)$. The parameter θ is not observed. It is never observed. This is the entire problem.

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Definition 1.2 (Signal Alphabet)**.** A *signal alphabet* is a finite set $\mathcal{Y}$ of observable behaviors. Conditional on θ, the observations $Y_1, Y_2, \ldots$ are i.i.d. with law P_θ on $\mathcal{Y}$. We work with the alphabet

$$\mathcal{Y} = \{\textsf{plan},\ \textsf{question},\ \textsf{normal},\ \textsf{haha},\ \textsf{read}\}, \tag{1}$$

where plan means she initiates a plan, question that she asks a follow-up question, normal an ordinary reply, haha a reply consisting only of "haha," and read that you were left on read.

Definition 1.3 (Likelihood Ratio and Evidence)**.** The *likelihood ratio* of a signal is $\Lambda(y) = P_1(y)/P_0(y)$, and its *evidence* is $\log_2 \Lambda(y)$, measured in bits. The cumulative log-likelihood ratio is $S_n = \sum_{k=1}^{n} \log \Lambda(Y_k)$. Its expected increment under P_1 is the Kullback–Leibler divergence $D_{\mathrm{KL}}(P_1 \,\|\, P_0)$ [3, 1].

Throughout we use the illustrative calibration below. Rows sum to one under each hypothesis.

Signal y	$P_1(y)$	$P_0(y)$	$\Lambda(y)$	Evidence (bits)
she initiates a plan	0.10	0.01	10.00	+3.32
asks a follow-up question	0.30	0.15	2.00	+1.00
replies normally	0.35	0.44	0.80	−0.33
replies "haha"	0.15	0.25	0.60	−0.74
leaves you on read	0.10	0.15	0.67	−0.58

Remark 1.4. A normal reply is evidence *against*. This is not a typo.

2 Identifiability

Definition 2.1 (Identifiability)**.** The parameter θ is *identifiable* from (Y_k) if $P_0 \neq P_1$, i.e. if the map $\theta \mapsto P_\theta^{\otimes \mathbb{N}}$ is injective [5].

Theorem 2.2 (Consistency)**.** *If θ is identifiable and $0 < \pi_0 < 1$, then $\mathbb{P}(\theta = 1 \mid \mathscr{F}_n) \to \mathbf{1}\{\theta = 1\}$ almost surely. If θ is not identifiable, then $\mathbb{P}(\theta = 1 \mid \mathscr{F}_n) = \pi_0$ for every n.*

Proof. By Bayes' rule the posterior log-odds are $\log \frac{\pi_0}{1-\pi_0} + S_n$. Under $\theta = 1$ the increments of S_n are i.i.d. with mean $D_{\mathrm{KL}}(P_1 \,\|\, P_0) > 0$, and under $\theta = 0$ with mean $-D_{\mathrm{KL}}(P_0 \,\|\, P_1) < 0$, both strictly signed by Gibbs' inequality when $P_0 \neq P_1$. The strong law of large numbers gives $S_n \to \pm\infty$ a.s. accordingly. If $P_0 = P_1$ then $\Lambda \equiv 1$ and $S_n \equiv 0$. □

Theorem 2.3 (She's Just Being Nice)**.** *Suppose she is* nice*, meaning that her signal law is the same for everyone regardless of interest: $P_1 = P_0$. Then no sequence of observations, of any length, changes the probability that she likes you. In particular, every inference you have drawn from her being nice to you is* measurable with respect to the trivial σ-algebra.

Proof. Immediate from Theorem 2.2. □

Remark 2.4. Theorem 2.3 explains a large fraction of the empirical literature on this question.

Example 2.5. Let $\pi_0 = 0.3$ and observe (normal, normal, haha, question, plan, normal). The total evidence is +2.59 bits, and the posterior is

$$\mathbb{P}(\theta = 1 \mid \mathscr{F}_6) = \left(1 + \frac{1-\pi_0}{\pi_0}\, e^{-S_6}\right)^{-1} \approx 0.72. \tag{2}$$

Note that almost all of this came from a single plan. Four of the six signals were evidence against.

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3 How Many Signals Do You Need?

You must eventually choose between two terminal actions: *ask her out* (correct if $\theta = 1$) and *move on* (correct if $\theta = 0$). Fix tolerated error rates $\alpha = \mathbb{P}_0(\text{ask})$ and $\beta = \mathbb{P}_1(\text{move on})$.

Definition 3.1 (Sequential Probability Ratio Test)**.** For thresholds $b < 0 < a$, the SPRT stops at $N = \inf\{n : S_n \notin (b, a)\}$ and asks her out if $S_N \geq a$, moves on if $S_N \leq b$.

Theorem 3.2 (Wald). *With $a = \log \frac{1-\beta}{\alpha}$ and $b = \log \frac{\beta}{1-\alpha}$, the SPRT has error rates at most (α, β) up to overshoot, and among all tests with these error rates it minimizes both $\mathbb{E}_0[N]$ and $\mathbb{E}_1[N]$. Moreover, ignoring overshoot,*

$$\mathbb{E}_1[N] \approx \frac{(1-\beta)\,a + \beta\,b}{D_{\mathrm{KL}}(P_1 \,\|\, P_0)}, \qquad \mathbb{E}_0[N] \approx \frac{(1-\alpha)\,b + \alpha\,a}{-D_{\mathrm{KL}}(P_0 \,\|\, P_1)}. \tag{3}$$

Proof. The error bounds follow from a change of measure on the stopping event, $\mathbb{P}_0(S_N \geq a) \leq e^{-a}\mathbb{P}_1(S_N \geq a)$, and its mirror image. Optimality is the Wald–Wolfowitz theorem [6]. For (3), apply Wald's identity $\mathbb{E}_\theta[S_N] = \mathbb{E}_\theta[N]\,\mathbb{E}_\theta[\log \Lambda(Y_1)]$ and approximate S_N by the boundary it crosses. □

Corollary 3.3 (The Eleven-Interaction Rule). *For the calibration of Section 1 and $\alpha = \beta = 0.05$: $a = -b = \log 19 \approx 2.94$, $D_{\mathrm{KL}}(P_1 \,\|\, P_0) \approx 0.241$ nats, $D_{\mathrm{KL}}(P_0 \,\|\, P_1) \approx 0.162$ nats, and*

$$\mathbb{E}_1[N] \approx 11.0, \qquad \mathbb{E}_0[N] \approx 16.3. \tag{4}$$

It takes about eleven interactions to learn that she likes you, and about sixteen to learn that she does not. Rejection is harder to detect because its signals are less distinctive.

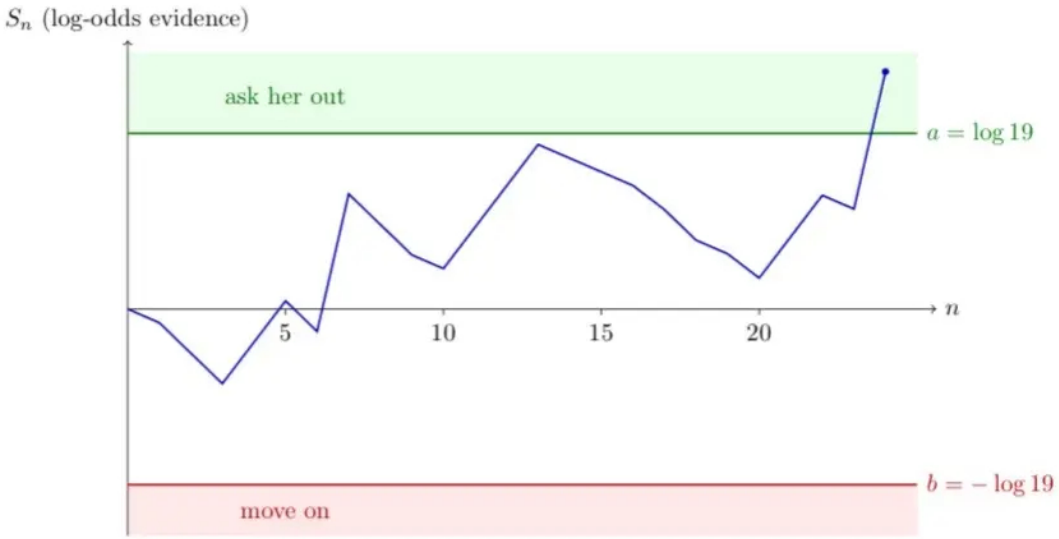


Figure 1: A sample path of S_n under $\theta = 1$. The walk nearly crossed at $n = 13$, then three ordinary replies pulled it back. The test terminated at $n = 24$ with a plan.

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4 Confirmation Bias and Almost-Sure Delusion

Definition 4.1 (κ-Biased Observer). A *κ-biased observer* with $\kappa \geq 1$ updates using the distorted increments

$$\tilde{\ell}(y) = \begin{cases} \kappa \log \Lambda(y), & \Lambda(y) > 1, \\ \log \Lambda(y), & \Lambda(y) \leq 1, \end{cases} \tag{5}$$

overweighting favorable signals by a factor κ and processing unfavorable ones correctly. This is a reduced form of the confirmatory bias of Rabin and Schrag [4].

Theorem 4.2 (Almost-Sure Delusion). *Let* $\kappa^* = -\mathbb{E}_0[\log\Lambda\,;\,\Lambda \le 1]/\mathbb{E}_0[\log\Lambda\,;\,\Lambda > 1]$. *If* $\kappa > \kappa^*$, *then under* $\theta = 0$ *the biased observer's posterior satisfies*

$$\tilde{\mathbb{P}}(\theta = 1 \mid \mathscr{F}_n) \longrightarrow 1 \qquad \mathbb{P}_0\text{-almost surely.} \tag{6}$$

For the calibration of Section 1, $\kappa^* \approx 2.28$.

Proof. Under $\mathbb{P}_0$ the distorted log-odds $\tilde{S}_n = \sum_{k\le n} \tilde{\ell}(Y_k)$ form a random walk with i.i.d. increments of mean $\mathbb{E}_0[\tilde{\ell}] = \kappa\,\mathbb{E}_0[\log\Lambda;\Lambda > 1] + \mathbb{E}_0[\log\Lambda;\Lambda \le 1]$, which is strictly positive iff $\kappa > \kappa^*$. By the strong law, $\tilde{S}_n \to +\infty$ a.s., so the distorted posterior tends to 1. Numerically, $\mathbb{E}_0[\log\Lambda;\Lambda > 1] = 0.01\log 10 + 0.15\log 2$ and $\mathbb{E}_0[\log\Lambda;\Lambda \le 1] = 0.44\log 0.795 + 0.25\log 0.6 + 0.15\log 0.667$. □

Corollary 4.3 (Delulu Is Not the Solulu). *An observer who counts each good sign as worth a little more than double its true weight will, with probability one, eventually become certain that she likes him, even when she does not. The data are not the problem. The likelihood function is.*

5 Time-Varying Interest: The Kalman Filter

Interest is not constant. Model it as a latent Gaussian process with linear observations:

$$x_t = \phi\, x_{t-1} + w_t, \quad w_t \sim N(0,q); \qquad y_t = x_t + v_t, \quad v_t \sim N(0,r), \tag{7}$$

where x_t is her interest at time t, y_t is what you think you observed, $|\phi| < 1$, and w_t, v_t are independent.

Theorem 5.1 (Irreducible Uncertainty). *The Kalman filter [2] for (7) has prior error variance* P_t *converging to the unique positive root of the discrete algebraic Riccati equation*

$$P = \phi^2 \frac{Pr}{P+r} + q \quad\Longleftrightarrow\quad P^2 + \big(r(1-\phi^2) - q\big)P - qr = 0, \tag{8}$$

and the steady-state posterior error variance $M = Pr/(P+r)$ *is strictly positive whenever* $q > 0$.

Proof. The Riccati recursion $P_{t+1} = \phi^2 P_t r/(P_t+r) + q$ is a monotone contraction on $(0,\infty)$ for $|\phi| < 1$, so it converges to its unique positive fixed point. Clearing denominators gives the quadratic, whose product of roots is $-qr < 0$, so exactly one root is positive. Then $M > 0$ since $P > 0$ and $r > 0$. □

Corollary 5.2 (You Will Never Be Sure). *With* $\phi = 0.95$, $q = 0.01$, $r = 0.25$: $P \approx 0.043$, *Kalman gain* $K = P/(P+r) \approx 0.15$, *and the best achievable posterior standard deviation is* $\sqrt{M} \approx 0.19$, *against a stationary standard deviation of* $\sqrt{q/(1-\phi^2)} \approx 0.32$. *Observing her forever reduces your uncertainty by at most* 40%. *The remaining* 60% *is structural.*

Remark 5.3. A gain of 0.15 means each new observation should move your estimate by 15% of the surprise. Most observers use a gain of approximately 1.

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6 The Optimal Experiment

Suppose that at any time you may run an alternative experiment, *asking her*, at cost c. Its outcome is fully revealing: $\Lambda(\text{yes}) = \infty$ and $\Lambda(\text{no}) = 0$.

Theorem 6.1 (Value of Information). *Let each passive interaction cost* $k > 0$ *(in time, overthinking, or screenshots sent to friends). Asking immediately dominates running the SPRT to completion whenever*

$$c \;<\; k\Big(\pi_0\,\mathbb{E}_1[N] + (1-\pi_0)\,\mathbb{E}_0[N]\Big) + L(\alpha,\beta), \tag{9}$$

where $L(\alpha,\beta) > 0$ *is the expected loss from the SPRT's residual errors.*

Proof. Asking costs c and has zero error. The SPRT costs $k\,\mathbb{E}[N]$ in expectation and errs with positive probability. Compare. □

Corollary 6.2 (Just Ask Her)**.** *For the calibration above, with $\pi_0 = 0.3$, the SPRT requires about* 14.7 *interactions in expectation and is still wrong* 5% *of the time. Asking requires one.*

7 Main Result

Theorem 7.1 (How to Tell If She Likes You)**.** *Let θ be latent interest and (Y_k) the observed signals.*

(a) If she is just being nice ($P_1 = P_0$), you cannot tell, ever (Theorem 2.3).

(b) Otherwise you can, and the optimal passive procedure needs about 11 *interactions if she does and* 16 *if she does not (Corollary 3.3).*

(c) If you overweight good signs by more than $\kappa^* \approx 2.28$, you will conclude that she does, almost surely, regardless of the truth (Theorem 4.2).

(d) If her interest changes over time, you will never be certain (Theorem 5.1).

(e) Asking her is a fully revealing experiment that dominates all of the above (Theorem 6.1).

Proof. Theorems 2.3, 3.2, 4.2, 5.1, and 6.1. □

8 Conclusion

We have shown that whether she likes you is a well-posed statistical question if and only if she is not just being nice, that the answer can be learned at a rate governed by a Kullback–Leibler divergence of about a quarter of a nat per interaction, that a sufficiently optimistic observer learns the wrong answer with probability one, and that the irreducible uncertainty of a time-varying signal can be computed from a Riccati equation. The dominant strategy is to ask. Future work will address the multi-observer setting (her friends also know, and they are better estimators than you) and the case in which she is also running an SPRT on you.

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